

Instructor: Date:

Learner: Regional Branch:

Program: Center:

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Let's Start!

These are the different domains for data warehouse.



Learning Outcome:

Designs the data structure based on the database structures and its characteristics.

Understand the concept of data warehouse, its components and characteristics.

Explain the different types of data warehouse, their characteristics and approaches

Can Do Statements:

- I can talk with a colleague about the domain where a warehouse can be implemented.
- I can describe a query-driven approach.
- I can explain update-driven approach.

Ask your partner:

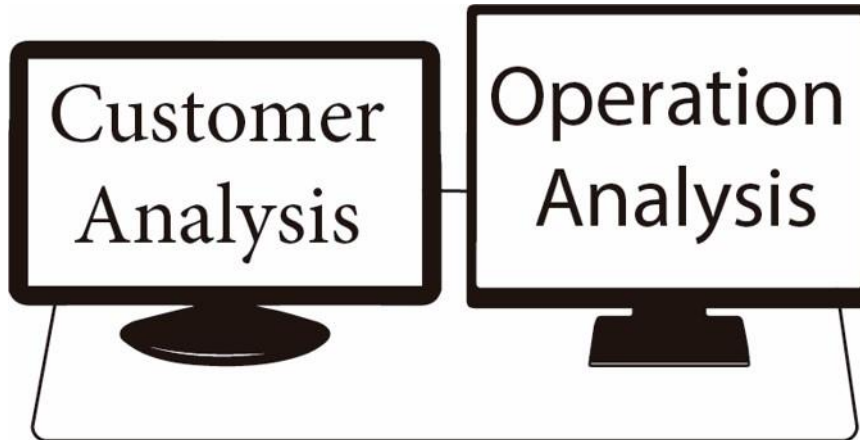
- How can data warehousing help in analysis of the operations of a business?
- What do you think is the most efficient approach to process information?





Let's Explore!

I won't use the query approach.



Instructions: Read the text below.

- Steven:** Caroline, what domains did you implement for the data warehouse?
- Caroline:** Well, I implemented data warehouse in tuning production strategies, customer analysis, and operations analysis.
- Steven:** What are they about?
- Caroline:** Tuning production strategies refer to the arrangement and management of products' portfolios; it is basically a comparison between sales in different periods of times. Customer analysis has got to do with customers' preferences when acquiring products. Operation analysis refers to customer relationship management and analyzing operations.
- Steven:** I see. You know, I have got to integrate some dissimilar databases, have you done it before?
- Caroline:** Yes, what approach have you thought about?
- Steven:** Approach? What do you mean by approach?
- Caroline:** Well, I am going to tell you what they are. There are two approaches: query-driven approach and update-driven approach. Did you hear about them before?
- Steven:** No, I didn't. What are they about?
- Caroline:** The query-driven approach is the usual way to integrate databases. This is basically a mediator. Some of its disadvantages are that you have got to do a complex filtering when using a query-driven approach. Also, it is very expensive.
- Steven:** Have you ever implemented a query-driven approach in a data warehouse?
- Caroline:** Actually, I haven't. You're going to have to use the update-driven approach since information is integrated in advance and it is always available for analysis. When I used it, I noticed a high fruition when copying, integrating, and processing data. The best of all is that processing data won't require an interface to prepare it at local roots.



Let's Practice!

Caroline describes what the query-driven means.

Instructions: Circle the correct option to complete the sentences.

1. Steven, you are (going / go / gone) to have to use the query-driven approach to integrate databases.
2. You (have got / had got / has got) to manage the query-driven approach as a mediator to combine databases.
3. Steven, you have got (doing / do / to do) a complex filtering when using a query-driven approach.
4. You are going (have / to have / had) to save money to get the query-driven approach. It is very expensive.
5. You have got to (use / uses / used) the query-driven approach as a way to filter important information.



Let's Practice!

Caroline explains to Steven what an update-driven approach is.

Instructions: Select the correct option to correctly complete each sentence.

1. (Have you ever implement / Have you ever implemented / Did you ever implement) the query-driven approach in data warehouse?

2. How (you integrate / have you integrated / did you integrate) the information?

3. What information (have you already had / have you had / did you already have) available for analysis?

4. (Did you used / Have you used / Did you use) the interface to process data?

5. Have (you achieved / did you achieved / did you achieve) good results when integrating data in the update-driven approach?



Extension Activity

Now it's your turn to compare.

Instructions: Complete the following steps. Fill in the Self-Assessment.

- Find information about the approaches to integrate heterogeneous databases.
- Create a Venn and Carroll diagram to sort out different and common features among the approaches.
- Give an oral presentation to your partners in order to explain the information stated in the diagram.

Self-Assessment

Yes

No

- I find information about the approaches to integrate heterogeneous databases.
- I classify different and common features among the approaches.
- I explain to my partners the information stated in the diagram.

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Target Words

- domain** (noun): an area to implement something
e.g. Caroline, what domains did you implement for the data warehouse?
- tuning** (noun): the appropriate adjustment of something
e.g. Well, I implemented data warehouse in tuning production strategies, customer analysis, and operations analysis.
- portfolio** (noun): the contents included in a specific file
e.g. Tuning production strategies refer to the arrangement and management of products' portfolios
- dissimilar** (adjective): that is heterogeneous
e.g. You know, I have got to integrate some dissimilar databases, have you done it before?
- fruition** (noun): to perform an action
e.g. When I used it, I noticed a high fruition when copying, integrating, and processing data.
- root** (noun): the origin of something
e.g. The best of all is that processing data won't require an interface to prepare it at local roots.



Learning Points

Present perfect vs Past simple - Interrogative form.

Both the **present perfect** and the **simple past** tenses are used to ask about situations that took place at some point in the past and may or may not have a relation with the present.

You can use the **present perfect** in questions to ask for information that started in the past and continues in the present. Use the simple past to ask for information that happened in the past.

To make a question in **present perfect**, you need to use the auxiliaries HAVE (I, you, we, they) and HAS (he, she, it) and a verb in past participle form:

Wh-question (optional) + Auxiliary Have/Has + Subject + Verb (in past participle) + Complement + ?

Examples:

Have you **used** the update-driven approach?

What approach **have** you **used** so far?

To make a question in **simple past**, you need to use the auxiliary *did* and a verb in base form:

Wh-question (optional) + Did + Subject + Verb (in base form) + Complement + ?

Examples:

Did you **use** tuning production strategies?

When **did** you **start** to work with a data warehouse?

Examples

- *Caroline, what domains did you implement for the data warehouse?*
- *You know, I have got to integrate some dissimilar databases, have you done it before?*
- *What approach have you thought about?*
- *There are two approaches: query-driven approach and update-driven approach. Did you hear about them before?*



Learning Points

Data Warehouse Approaches

There are different ways to work with **data warehouse**. Some of these ways or approaches to work with **data warehouse** support the work done by executives.

The **Query-Driven approach** is the most popular approach to combine heterogeneous databases. This approach is being used as integrator or mediator. Some of its characteristics are:

- Queries are mapped and sent to the local query processor.
- The results are integrated into a global answer set.

Its disadvantages are:

- It needs complex combination and filtering.
- It's not an efficient approach.
- It's very expensive.

The **Update-Driven approach** is an alternative approach. Its main feature is that the information from different databases are combined beforehand and saved in the warehouse. This information is always available for analysis.

Its advantages are:

- It provides a high performance.
- The data is processed and integrated in advance.

Examples

- *The query-driven approach is the usual way to integrate databases. This is basically a mediator. Some of its disadvantages are that you have got to do a complex filtering when using a query-driven approach. Also, it is very expensive.*
- *You're going to have to use the update-driven approach since information is integrated in advance and it is always available for analysis. When I used it, I noticed a high fruition when copying, integrating, and processing data. The best of all is that processing data won't require an interface to prepare it at local roots.*

